Haiqing Gao

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Research Interests

My research uses mathematical modeling and stochastic optimization to improve nonprofit operations. I study structural properties of optimal allocation and routing policies to connect optimization theory with interpretable AI for online sequential decision-making. More broadly, I develop stochastic models that yield actionable managerial insights and interpretable decisions.

Education

Northwestern University
Ph.D. Candidate in Industrial Engineering and Management Sciences (IEMS)
Sep. 2022 – Expected Jun. 2027

Advisor: Prof. Seyed M.R. Iravani
Ph.D. Dissertation: *Sequential Fair Allocation and Routing in Nonprofit Operations*

This dissertation develops mathematical models for fairness-aware sequential allocation and routing under uncertainty, motivated by nonprofit operations such as food banks and humanitarian relief systems. Using stochastic optimization, dynamic programming, probability, and convex analysis, it derives structural properties of optimal policies that lead to interpretable and implementable decision rules. The dissertation establishes threshold allocation policies, characterizes when demand uncertainty determines the optimal visitation order, clarifies that fairness objectives need not align with fairness metrics, and shows that optimizing routing decisions can substantially improve fairness metrics even when fairness objective value gains are modest.

The Chinese University of Hong Kong, Shenzhen
M.Phil. in Computer Information and Engineering Sep. 2020 – Jun. 2022
B.S. in Applied Mathematics (First Honor) Sep. 2016 – May 2020

Research Experience

Northwestern University | Ph.D. Research
Concept Drift Detection during Online Federated Learning from Hybrid Data May 2026 – Present

- Proved a dynamic regret bound of $O(\sqrt{T}) + \Delta_{[1,T]}$ for the *FedHD* gradient tracking algorithm (Gao et al., 2023) under nonstationary conditions.
- Constructed an optimistic loss estimator for drift detection satisfying the requirements of the Master-FL framework *(Ganguly and Aggarwal, 2024)*.
- Extended *Ganguly and Aggarwal's (2024)* worst-case analysis to establish a probabilistic bound.

Sequential Fair Allocation and Collection of a Capacitated Truck with Stochastic Supply in Nonprofit Operations Apr. 2026 – Present

- Developed dynamic programs for allocation, collection, and joint allocation and collection by a capacitated truck with stochastic supply nodes.

- Established monotonicity and threshold structures for optimal allocation and collection policies, including thresholds in demand and prior minimum fill or collection rates, and proved monotonicity of the corresponding thresholds.
- Designed interpretable heuristics for the joint allocation and collection model using the threshold structure and equating property of the optimal policy.

Multi-vehicle Sequential Fair Allocation and Routing: A Linear Approximation Model

Jan. – Mar. 2026

- Formulated a stochastic mixed-integer program for multi-vehicle sequential fair allocation and routing.
- Developed a two-stage stochastic approximation that separates first-stage clustering from second-stage sequential allocation and routing.
- Evaluated the approximation numerically and found a gap of less than 0.01 in the fairness objective value relative to the exact optimum.

Sequential Fair Allocation and Routing in Nonprofit Operations Jan. 2023 – Jun. 2026

- Formulated dynamic programs incorporating routing decisions under two prevailing fairness objectives: *ex-post* and *forward* fairness.
- Established demand thresholds and an equating property to guide heuristic design for both objectives.
- Proved that routing in decreasing order of the coefficient of variation is optimal under specified conditions for the *ex-post* objective; numerical experiments revealed no consistent ordering for the *forward* objective.
- Demonstrated numerically that choosing an unsuitable route can reduce fairness metrics by 10%.
- Identified differences of up to 5% in short- and long-term fairness metrics between *ex-post* and *forward* objectives, informing objective selection based on managerial priorities.
- Proposed the PPA-deCV heuristic, which performed within 3% of the optimal allocation and routing policy under both objectives.
- Mentored a first-year Ph.D. student in simulation and experimental design.
- Developed a manuscript submitted to *Operations Research* and currently under review.

CUHK-Shenzhen | Master's Research

Hybrid Machine Learning via Gradient Tracking May 2020 – Jun. 2022

- Designed a federated learning algorithm using gradient tracking to learn from hybrid data.
- Proved a theoretical bound $O(1/T)$ on the average gradient.
- Improved accuracy by 50% and achieved twice the convergence speed of the local training model.
- Published the resulting work in the *Journal of the Franklin Institute*.

Federated Q-learning for mHealth Interventions Oct. 2020 – Dec. 2020

- Developed a personalized federated Q-learning algorithm incorporating Bayesian learning.
- Achieved threefold faster convergence while maintaining accuracy through decentralized data processing.
- Collaborated with a first-year Ph.D. student to implement the algorithm and simulate mobile health data.

CUHK-Shenzhen | Research Assistant Mar. 2019 – Aug. 2019

Fair Driver Order Dispatch on the Online Ride-Hailing Platform

- Formulated **a** stochastic online matching problem with customer abandonment and reusable drivers.
- Applied a multi-armed bandit approach and developed an online Hungarian bottleneck algorithm to improve matching fairness and efficiency.
- Mentored a junior student in algorithm implementation and simulation.

Publications & Working Papers

Gao, H., S. M. R. Iravani, and S. R. Sinclair. *Sequential Fair Allocation and Routing in Nonprofit Operations.* Submitted to *Operations Research*, June 2026.

Gao, H., S. Ge, and T.-H. Chang. *FedHD: Communication-Efficient Federated Learning from Hybrid Data. Journal of the Franklin Institute*, 2023.

Gao, H., and S. M. R. Iravani. *Sequential Fair Allocation and Collection of a Capacitated Truck with Stochastic Supply in Nonprofit Operations*. Working paper.

Gao, H., and S. Mehrotra. *Concept Drift Detection during Online Federated Learning from Hybrid Data*. Working paper.

Presentations

History Dependent or Memoryless? Objective Selection for Fair Sequential Allocation and Routing in Non-profit Operations

- 2025 INFORMS Annual Meeting — Oct. 2025
  Invited Talk for Public Sector OR (Session Chair)
- 3rd Annual Supply Chain and Operations Management Conference (SCOM) — Aug. 2025
  Invited Talk for Dedicated Student Session

Fair Allocation in Nonprofit Operations: Optimal Routing Policy

- Chicago Area Society for Industrial and Applied Mathematics (SIAM) Student Conference 2025 (CASSC25) — Mar. 2025
  Poster & Lightning Talk
- Toyota Technological Institute at Chicago (TTIC) Workshop — Aug. 2024
  Poster & Lightning Talk

Federated Q-learning in Mobile Health (mHealth) Intervention

- Reinforcement Learning Project Presentations for Graduate Students — Nov. 2020
  Poster & Lightning Talk

Fair Order Dispatch on the Online Ride-Hailing Platform

- Shenzhen Research Institute of Big Data Research Presentation — Jul. 2019

Honors

Northwestern University

Dissertation Final Year Fellowship from Transportation Center — 2026 – 2027
Institute of Industrial & Systems Engineers (IISE)
Future Faculty Fellows Program (Top 10%) — 2025 – 2026

The Chinese University of Hong Kong, Shenzhen

Master's List (Top 10%) — 2017 – 2019
Dean's List (Top 10%) — 2016 – 2019
Academic Performance Scholarship (Top 7%) — 2017

Teaching Experience

Lecturer

Northwestern University

Graduate Boot Camp Instructor for Python Sep. 2026

- Designed a six-hour boot camp comprising four hours of lectures and two hours of office hours for first-year graduate students.
- Taught core Python packages and data structures and guided students through installing Anaconda and gurobipy.
- Demonstrated AI-assisted coding and facilitated a group discussion.

Graduate Boot Camp Instructor for Probability Sep. 2025

- Delivered six hours of lectures for first-year graduate students, covering fundamental probability concepts.
- Introduced measure theory, convergence theory, and limit theorems.

Teaching Assistant

- Designed materials for lab sessions and led sessions to reinforce course concepts.
- Held office hours and directed students to relevant learning resources.
- Graded homework, quizzes, and exams and provided constructive feedback to students.

Northwestern University

IEMS 365: Analytics for Social Good Winter/Spring Quarters 2025/2026

- Supported course project design by running experiments to validate solutions and providing feedback to the instructor.
- Met with student groups to review project progress and provide feedback on their approaches and results.

IEMS 302: Probability Winter/Fall Quarters 2024/2025/2026

- Delivered three lectures on behalf of the instructor, covering combinatorics, conditional probability, continuous distributions, and joint distributions.
- Developed homework assignments, quizzes, and exams and prepared solutions.

IEMS 303: Statistics Fall Quarter 2024

IEMS 307: Quality Improvement by Experimental Design Spring Quarter 2024

IEMS 201: Introduction to Statistics or Data-Driven Decision-Making Fall Quarter 2023

The Chinese University of Hong Kong, Shenzhen

MATH 3253: Complex Variables Spring Semester 2022

STA 2001: Probability and Statistics I Spring Semester 2019

Leadership & Service

The Graduate School Diversity Peer Mentor | *Northwestern University* Dec. 2023 – Apr. 2026

- Guided first-year international Ph.D. students through academic and cultural transitions.

INFORMS Peer Mentor | *Northwestern University INFORMS Student Chapter* Dec. 2023 – Aug. 2025

- Mentored two incoming IEMS Ph.D. students during their transition to graduate study.

English Language Program Foundations Peer Mentor | *Northwestern University*

Jun. – Aug. 2024 and 2025

- Led weekly adjustment meetings for 60 incoming international students and served as a mock student in teaching assessments.

McCormick Graduate Leadership Council Navigating First Year Panelist | *Northwestern University*
Oct. 2024

- Shared graduate school experiences and resources with first-year students.

INFORMS Annual Meeting Session Chair | *Public Sector OR* Apr. – Oct. 2024

- Recruited speakers and chaired a session on resource allocation in nonprofit operations.

INFORMS Student Chapter Organizer | *Northwestern University* Jan.–Mar. 2024

- Organized interdisciplinary seminars and a networking event for more than 70 students.

Skills

Programming Languages/Packages: Python, MATLAB, R, LaTeX, PyTorch, Gurobipy
Languages: Chinese (Native), English (Proficient)

References

Professor Seyed M.R. Iravani

- Professor of Industrial Engineering and Management Sciences, McCormick School of Engineering; Professor of Operations Management (Courtesy), Kellogg School of Management, Northwestern University
- Technological Institute, 2145 Sheridan Road, E284, Evanston, IL 60208-3109
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Professor Sanjay Mehrotra

- Emma Ann Reynolds Professor of Industrial Engineering and Management Sciences, McCormick School of Engineering, Northwestern University; Director, Center for Engineering and Health; Co-Director, Project Minerva
- Technological Institute, 2145 Sheridan Road, C246, Evanston, IL 60208-3109
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Professor Jill H. Wilson

- Professor of Instruction of Industrial Engineering and Management Sciences; Associate Chair for Undergraduate Education, McCormick School of Engineering, Northwestern University
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- Phone: (847) 467-1551

Professor Izzy Grosof

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